

Table 1: Pass-sum execution time (s, median per iteration; sum of pass medians, not full executable wall time) and speedup split by pass type. When present, the OctTree row includes ColorOctree passes; a separate ColorOctree row reports only those passes. ADT fields = total fields in the selected benchmark ADT (prefer parsed ADT definition; fall back to `@BENCH adt_fields`). SoA bufs = 1 tag buffer plus one buffer for each scalar field, plus one buffer for each non-self packed field annotated Linear, plus recursively counted buffers for each non-self packed field annotated Factored; self-recursive fields add no new buffers. Speedup $>1\times$ means the denominator is faster; **bold** marks $>1.1\times$.

Program	ADT	SoA	End-to-end			Fold passes			Map passes		
	fields	bufs	Am (s)	Sm (s)	Am/Sm	Am (s)	Sm (s)	Am/Sm	Am (s)	Sm (s)	Am/Sm
OctTree	16	9	0.6771	0.3733	1.81 $\times$	0.3730	0.3224	1.16 $\times$	0.3041	0.0510	5.97 $\times$
Compiler	9	8	0.2904	0.0607	4.78 $\times$	0.1308	0.0395	3.31 $\times$	0.1596	0.0212	7.52 $\times$
DBQuery	15	13	0.2560	0.1111	2.30 $\times$	0.1191	0.0877	1.36 $\times$	0.1369	0.0234	5.85 $\times$
DecisionTree	7	6	3.766	2.985	1.26 $\times$	3.766	2.985	1.26 $\times$	–	–	–
DomTree	15	14	0.2017	0.1161	1.74 $\times$	0.1427	0.0595	2.40 $\times$	0.0590	0.0566	1.04 $\times$
KDTree	17	16	0.1823	0.1392	1.31 $\times$	0.1823	0.1392	1.31 $\times$	–	–	–
LinearListReduction	11	11	0.0306	0.0052	5.84 $\times$	0.0306	0.0052	5.84 $\times$	–	–	–
reduceNestedList	3	3	0.0871	0.0074	11.82 $\times$	0.0871	0.0074	11.82 $\times$	–	–	–
List	2	2	0.5768	0.2914	1.98 $\times$	0.3252	0.1100	2.96 $\times$	0.2516	0.1814	1.39 $\times$
MonoTree	3	2	0.0608	0.0373	1.63 $\times$	0.0173	0.0220	0.79 $\times$	0.0435	0.0154	2.83 $\times$
ObjectGraph	5	4	0.2397	0.1429	1.68 $\times$	0.0973	0.0806	1.21 $\times$	0.1424	0.0624	2.28 $\times$
PiecewiseFunctions	8	7	0.3832	0.1383	2.77 $\times$	0.1449	0.0917	1.58 $\times$	0.2383	0.0465	5.12 $\times$
TernaryTree	5	3	0.1111	0.0696	1.60 $\times$	0.0292	0.0289	1.01 $\times$	0.0819	0.0407	2.01 $\times$
Trie	10	9	0.2148	0.0716	3.00 $\times$	0.0664	0.0323	2.05 $\times$	0.1484	0.0393	3.78 $\times$
ColorOctree	25	18	0.0903	0.0726	1.24 $\times$	0.0903	0.0726	1.24 $\times$	–	–	–

Table 2: Per-pass performance for **OctTree**, OctTree SoA uses 9 buffers; ColorOctree SoA uses 18 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
barnesHutPotential	F	11/16	31%	0.0417 $\pm$ 0.00226	0.0390 $\pm$ 0.00302	1.07 $\times$
countActive	F	10/16	38%	0.0375 $\pm$ 0.00003	0.0289$\pm$0.00002	1.30 $\times$
countParticles	F	8/16	50%	0.0354 $\pm$ 0.00003	0.0274$\pm$0.00003	1.29 $\times$
fmmPotential	F	12/16	25%	0.0969 $\pm$ 0.00005	0.0953 $\pm$ 0.00002	1.02 $\times$
sumEnergy	F	12/16	25%	0.0356 $\pm$ 0.00008	0.0319$\pm$0.00001	1.12 $\times$
sumMass	F	10/16	38%	0.0357 $\pm$ 0.00003	0.0272$\pm$0.00001	1.31 $\times$
clearFlags	M	15/16	6%	0.1534 $\pm$ 0.00022	0.0064$\pm$0.00001	23.99 $\times$
scaleEnergy	M	16/16	0%	0.1507 $\pm$ 0.00020	0.0446$\pm$0.00026	3.38 $\times$
paletteEntriesQuantized	F	13/25	48%	0.0438 $\pm$ 0.00003	0.0299$\pm$0.00006	1.46 $\times$
quantizationErrorProxy	F	10/25	60%	0.0464 $\pm$ 0.00010	0.0426 $\pm$ 0.00002	1.09 $\times$
Total				0.6771	0.3733	1.81 $\times$
Geomean						1.79 $\times$

Table 3: Per-pass performance for **Compiler**, ADT has 9 fields, SoA uses 8 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
blockCountPass	F	2/9	78%	0.0125 $\pm$ 0.00000	0.0019$\pm$0.00000	6.54 $\times$
branchStatsPass	F	2/9	78%	0.0121 $\pm$ 0.00001	0.0036$\pm$0.00000	3.39 $\times$
castInstCountPass	F	3/9	67%	0.0190 $\pm$ 0.00000	0.0013$\pm$0.00000	14.80 $\times$
has cycle	F	4/9	56%	0.0245 $\pm$ 0.00022	0.0176$\pm$0.00029	1.39 $\times$
instCountPass	F	2/9	78%	0.0125 $\pm$ 0.00002	0.0019$\pm$0.00000	6.54 $\times$
latencyModelPass	F	3/9	67%	0.0125 $\pm$ 0.00002	0.0030$\pm$0.00000	4.18 $\times$
memoryOpStatsPass	F	3/9	67%	0.0123 $\pm$ 0.00001	0.0038$\pm$0.00000	3.22 $\times$
throughputModelPass	F	3/9	67%	0.0125 $\pm$ 0.00001	0.0030$\pm$0.00000	4.17 $\times$
verifyIR	F	9/9	0%	0.0129 $\pm$ 0.00001	0.0034$\pm$0.00000	3.74 $\times$
stripSideEffectsPass	M	7/9	22%	0.0802 $\pm$ 0.00018	0.0103$\pm$0.00001	7.78 $\times$
targetReturnPass	M	9/9	0%	0.0794 $\pm$ 0.00037	0.0109$\pm$0.00001	7.27 $\times$
Total				0.2904	0.0607	4.78 $\times$
Geomean						4.86 $\times$

Table 4: Per-pass performance for **DBQuery**, ADT has 15 fields, SoA uses 13 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
countJoins	F	3/15	80%	0.0185 $\pm$ 0.00001	0.0112$\pm$0.00001	1.65 $\times$
filterSelectivitySkew	F	5/15	67%	0.0184 $\pm$ 0.00001	0.0123$\pm$0.00001	1.49 $\times$
hashJoinPressure	F	5/15	67%	0.0196 $\pm$ 0.00001	0.0125$\pm$0.00002	1.56 $\times$
sumCost	F	6/15	60%	0.0185 $\pm$ 0.00001	0.0149$\pm$0.00003	1.24 $\times$
sumMemory	F	6/15	60%	0.0183 $\pm$ 0.00001	0.0147$\pm$0.00002	1.24 $\times$
sumRows	F	6/15	60%	0.0260 $\pm$ 0.00002	0.0220$\pm$0.00002	1.18 $\times$
clearQueryFlags	M	14/15	7%	0.0684 $\pm$ 0.00014	0.0085$\pm$0.00001	8.09 $\times$
scaleCosts	M	15/15	0%	0.0685 $\pm$ 0.00238	0.0150$\pm$0.00004	4.58 $\times$
Total				0.2560	0.1111	2.30 $\times$
Geomean						2.00 $\times$

Table 5: Per-pass performance for **DecisionTree**, ADT has 7 fields, SoA uses 6 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
classify Batch	F	5/7	29%	3.232 $\pm$ 0.0044	2.638$\pm$0.0084	1.22 $\times$
classify Depth	F	4/7	43%	0.0463 $\pm$ 0.00003	0.0370$\pm$0.00006	1.25 $\times$
classify tree	F	5/7	29%	0.0463 $\pm$ 0.00001	0.0371$\pm$0.00008	1.25 $\times$
countFeatureUses	F	3/7	57%	0.0493 $\pm$ 0.00002	0.0321$\pm$0.00003	1.54 $\times$
countLeaves	F	2/7	71%	0.0470 $\pm$ 0.00032	0.0242$\pm$0.00002	1.94 $\times$
countNodes	F	2/7	71%	0.0471 $\pm$ 0.00005	0.0248$\pm$0.00001	1.90 $\times$
countSmallLeaves	F	3/7	57%	0.0501 $\pm$ 0.00002	0.0324$\pm$0.00004	1.55 $\times$
inferenceCost	F	2/7	71%	0.0499 $\pm$ 0.00002	0.0306$\pm$0.00003	1.63 $\times$
sumImpurity	F	3/7	57%	0.0494 $\pm$ 0.00158	0.0326$\pm$0.00002	1.52 $\times$
sumPathLengths	F	3/7	57%	0.0516 $\pm$ 0.00002	0.0340$\pm$0.00002	1.52 $\times$
sumSamples	F	3/7	57%	0.0473 $\pm$ 0.00002	0.0320$\pm$0.00154	1.48 $\times$
treeDepth	F	2/7	71%	0.0503 $\pm$ 0.00003	0.0301$\pm$0.00001	1.67 $\times$
Total				3.766	2.985	1.26 $\times$
Geomean						1.52 $\times$

Table 6: Per-pass performance for **DomTree**, ADT has 15 fields, SoA uses 14 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
count styled	F	3/15	80%	0.0351 $\pm$ 0.00110	0.0116$\pm$0.00003	3.03 $\times$
find max Bottom	F	5/15	67%	0.0358 $\pm$ 0.00014	0.0171$\pm$0.00009	2.09 $\times$
SumArea	F	6/15	60%	0.0368 $\pm$ 0.00006	0.0192$\pm$0.00004	1.92 $\times$
sumTextWidth	F	3/15	80%	0.0351 $\pm$ 0.00013	0.0116$\pm$0.00005	3.02 $\times$
computeWidths	M	14/15	7%	0.0301$\pm$0.00007	0.0450 $\pm$ 0.00007	0.67 $\times$
scaleLayout	M	15/15	0%	0.0289 $\pm$ 0.00006	0.0116$\pm$0.00002	2.48 $\times$
Total				0.2017	0.1161	1.74 $\times$
Geomean						1.98 $\times$

Table 7: Per-pass performance for **KDTree**, ADT has 17 fields, SoA uses 16 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
countInRange tight_box	F	13/17	24%	0.0283 $\pm$ 0.00004	0.0200$\pm$0.00002	1.41 $\times$
Find nearest Neighbour	F	13/17	24%	0.0285 $\pm$ 0.00172	0.0212$\pm$0.00244	1.35 $\times$
photonMappingBenchmark	F	12/17	29%	0.0429 $\pm$ 0.00005	0.0406 $\pm$ 0.00002	1.06 $\times$
pointCloudNeighborhood	F	11/17	35%	0.0256 $\pm$ 0.00004	0.0166$\pm$0.00001	1.55 $\times$
sumMassInRange	F	14/17	18%	0.0279 $\pm$ 0.00003	0.0202$\pm$0.00002	1.38 $\times$
twoPointCorrelation bin_8_16	F	11/17	35%	0.0292 $\pm$ 0.00004	0.0207$\pm$0.00002	1.41 $\times$
Total				0.1823	0.1392	1.31 $\times$
Geomean						1.35 $\times$

Table 8: Per-pass performance for **LinearListReduction**, ADT has 11 fields, SoA uses 11 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
reduction	F	2/11	82%	0.0306 $\pm$ 0.00024	0.0052$\pm$0.00002	5.84 $\times$
Total				0.0306	0.0052	5.84 $\times$
Geomean						5.84 $\times$

Table 9: Per-pass performance for **reduceNestedList**, ADT has 3 fields, SoA uses 3 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
reduction	F	2/3	33%	0.0871 $\pm$ 0.00032	0.0074$\pm$0.00001	11.82 $\times$
Total				0.0871	0.0074	11.82 $\times$
Geomean						11.82 $\times$

Table 10: Per-pass performance for **List**, ADT has 2 fields, SoA uses 2 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
length List	F	1/2	50%	0.0476 $\pm$ 0.00005	0.0215$\pm$0.00001	2.22 $\times$
sumList	F	2/2	0%	0.1388 $\pm$ 0.00006	0.0434$\pm$0.00002	3.19 $\times$
sumList tail recursive	F	2/2	0%	0.1388 $\pm$ 0.00004	0.0451$\pm$0.00002	3.08 $\times$
add1 List	M	2/2	0%	0.2516 $\pm$ 0.00288	0.1814$\pm$0.00044	1.39 $\times$
Total				0.5768	0.2914	1.98 $\times$
Geomean						2.34 $\times$

Table 11: Per-pass performance for **MonoTree**, ADT has 3 fields, SoA uses 2 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
sumTree	F	3/3	0%	0.0089$\pm$0.00120	0.0111 $\pm$ 0.00003	0.80 $\times$
sumTree TailRec	F	3/3	0%	0.0084$\pm$0.00001	0.0109 $\pm$ 0.00003	0.77 $\times$
add1Tree	M	3/3	0%	0.0435 $\pm$ 0.00008	0.0154$\pm$0.00123	2.83 $\times$
Total				0.0608	0.0373	1.63 $\times$
Geomean						1.21 $\times$

Table 12: Per-pass performance for **ObjectGraph**, ADT has 5 fields, SoA uses 4 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across –iterate runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
countLargeItems	F	3/5	40%	0.0136 $\pm$ 0.00001	0.0104$\pm$0.00003	1.30 $\times$
countMarked	F	3/5	40%	0.0135 $\pm$ 0.00001	0.0109$\pm$0.00003	1.24 $\times$
countSurvivors	F	4/5	20%	0.0145 $\pm$ 0.00001	0.0132 $\pm$ 0.00001	1.10 $\times$
deadBytes	F	4/5	20%	0.0149 $\pm$ 0.00126	0.0127$\pm$0.00000	1.17 $\times$
liveBytes	F	4/5	20%	0.0141 $\pm$ 0.00117	0.0127$\pm$0.00001	1.11 $\times$
sumObjIds	F	3/5	40%	0.0135 $\pm$ 0.00001	0.0104$\pm$0.00003	1.29 $\times$
totalHeapSize	F	3/5	40%	0.0133 $\pm$ 0.00002	0.0102$\pm$0.00017	1.30 $\times$
sweepUnmarked	M	5/5	0%	0.0709 $\pm$ 0.00015	0.0302$\pm$0.00008	2.35 $\times$
touchHotObjects	M	5/5	0%	0.0716 $\pm$ 0.00010	0.0321$\pm$0.00002	2.23 $\times$
Total				0.2397	0.1429	1.68 $\times$
Geomean						1.40 $\times$

Table 13: Per-pass performance for `PiecewiseFunctions`, ADT has 8 fields, SoA uses 7 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across $-iterate$ runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
autorefineMaxLevel	F	4/8	50%	0.0226 $\pm$ 0.00001	0.0195$\pm$0.00004	1.16$\times$
compressMass	F	3/8	62%	0.0204 $\pm$ 0.00095	0.0117$\pm$0.00121	1.74$\times$
lbDeuxLoadProxy	F	5/8	38%	0.0225 $\pm$ 0.00001	0.0183$\pm$0.00003	1.23$\times$
norm2Estimate	F	5/8	38%	0.0258 $\pm$ 0.00004	0.0179$\pm$0.00003	1.44$\times$
pmapCutHistogram	F	4/8	50%	0.0328 $\pm$ 0.00002	0.0128$\pm$0.00001	2.57$\times$
truncateTolViolations	F	3/8	62%	0.0208 $\pm$ 0.00001	0.0116$\pm$0.00002	1.80$\times$
addConstPW	M	8/8	0%	0.1195 $\pm$ 0.00024	0.0156$\pm$0.00002	7.68$\times$
diffPW	M	8/8	0%	0.1189 $\pm$ 0.00228	0.0310$\pm$0.00003	3.84$\times$
Total				0.3832	0.1383	2.77$\times$
Geomean						2.17$\times$

Table 14: Per-pass performance for `TernaryTree`, ADT has 5 fields, SoA uses 3 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across $-iterate$ runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
sum tree	F	5/5	0%	0.0292 $\pm$ 0.00010	0.0289 $\pm$ 0.00003	1.01 $\times$
add 1 tree	M	5/5	0%	0.0819 $\pm$ 0.00180	0.0407$\pm$0.00009	2.01$\times$
Total				0.1111	0.0696	1.60$\times$
Geomean						1.43$\times$

Table 15: Per-pass performance for `Trie`, ADT has 10 fields, SoA uses 9 buffers. Times are median per iteration (s); $\pm$ shows standard error of the mean across $-iterate$ runs. T: F=fold, M=map. Uses: fields accessed / total (recursive + non-recursive). Dead%: fraction of fields not accessed by this pass. Speedup $>1\times$ means SoA is faster. OOM = out of memory.

Pass	T	Uses	Dead%	AoS med $\pm$ err	SoA med $\pm$ err	Speedup
autocompleteTopKProxy	F	5/10	50%	0.0151 $\pm$ 0.00001	0.0083$\pm$0.00004	1.81$\times$
countLazyNodes	F	4/10	60%	0.0133 $\pm$ 0.00001	0.0067$\pm$0.00001	2.00$\times$
countTerminals	F	3/10	70%	0.0126 $\pm$ 0.00002	0.0059$\pm$0.00001	2.15$\times$
sumPrefixFreq	F	3/10	70%	0.0127 $\pm$ 0.00002	0.0057$\pm$0.00001	2.22$\times$
sumSubtreeHints	F	3/10	70%	0.0127 $\pm$ 0.00001	0.0057$\pm$0.00001	2.21$\times$
decayTrieStats	M	10/10	0%	0.0741 $\pm$ 0.00012	0.0242$\pm$0.00003	3.06$\times$
resetTraversalState	M	10/10	0%	0.0743 $\pm$ 0.00013	0.0151$\pm$0.00002	4.93$\times$
Total				0.2148	0.0716	3.00$\times$
Geomean						2.48$\times$